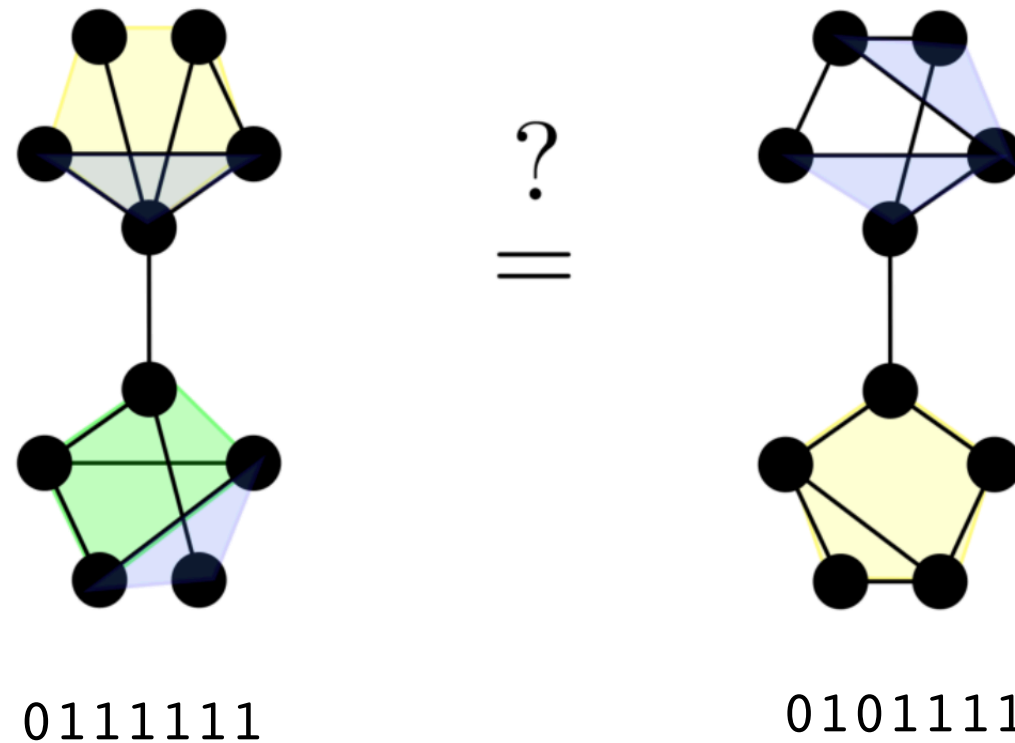


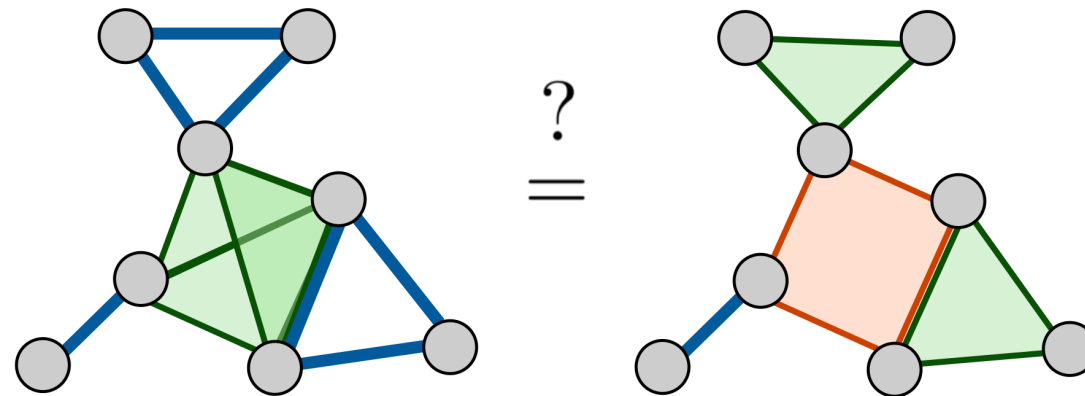
Information theory for hypergraph similarity



Challenges in hypergraph comparison

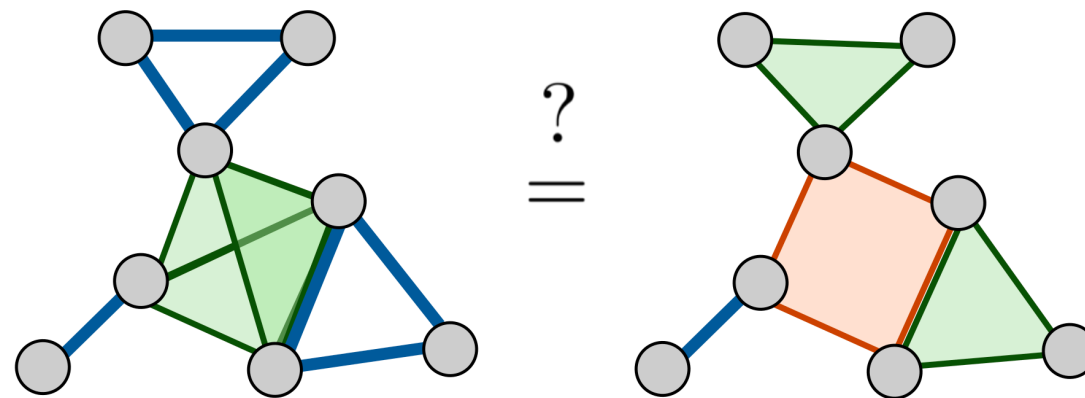
Challenges in hypergraph comparison

How to quantify similarity across orders?

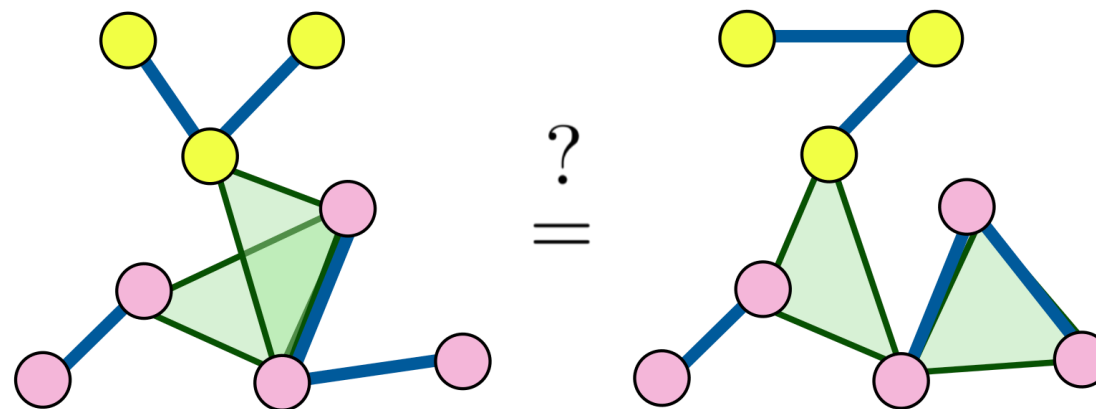


Challenges in hypergraph comparison

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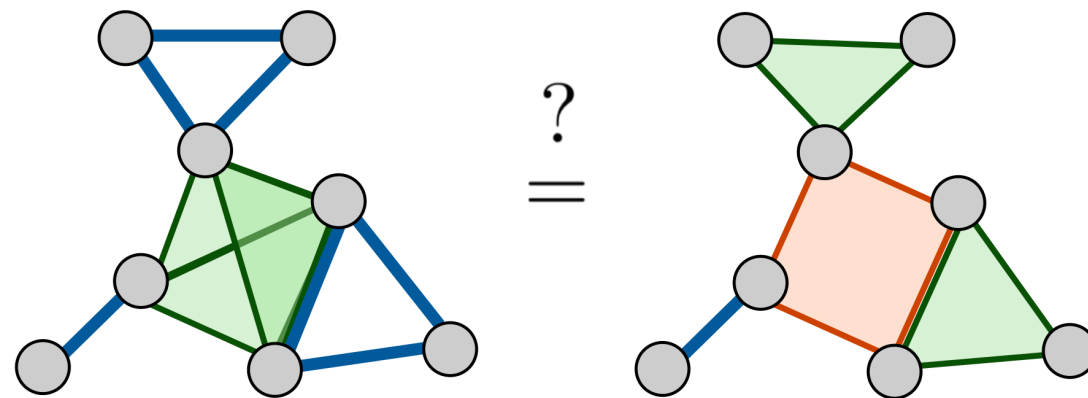


How to assess similarity at multiple scales?

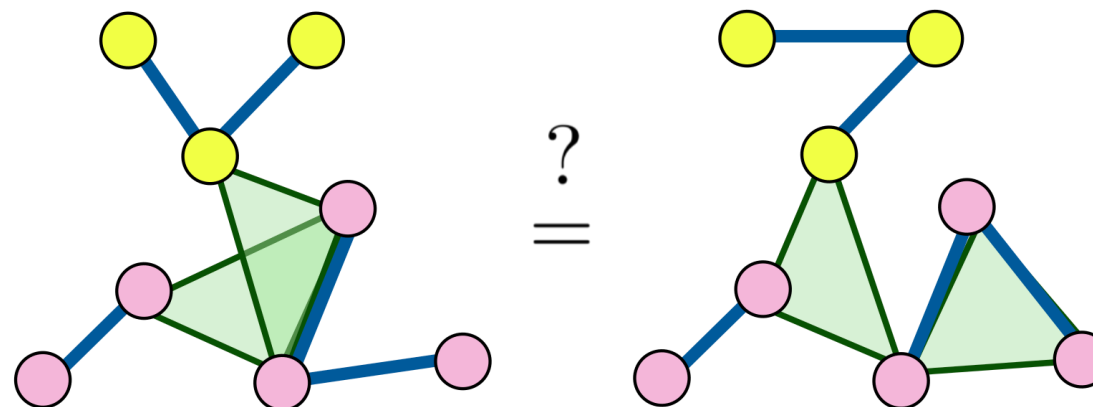


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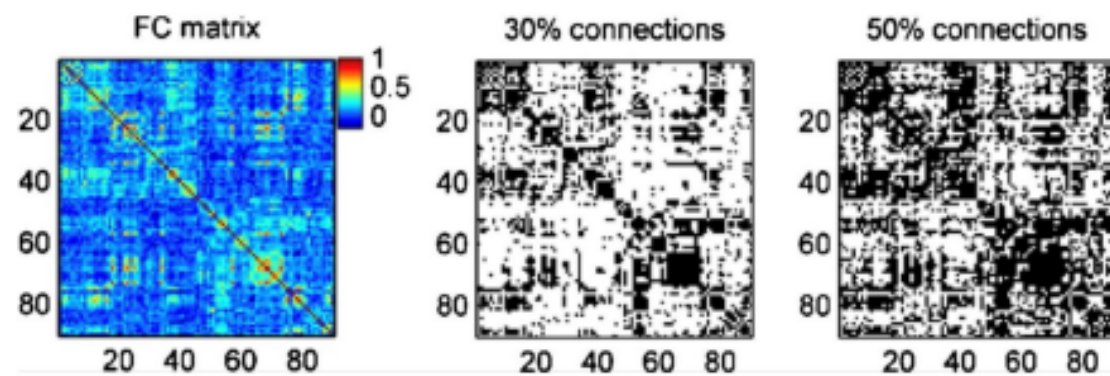
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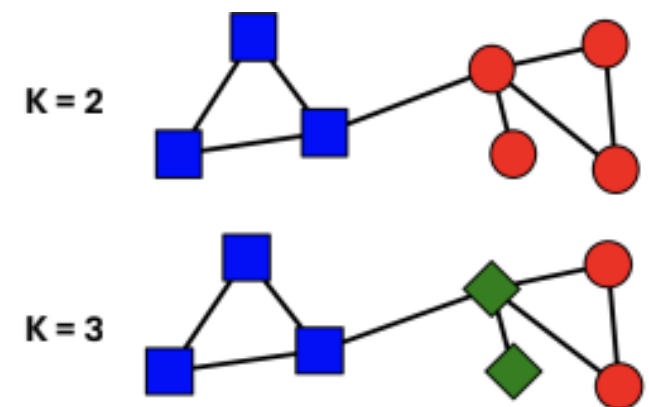
How to correct for chance and avoid spuriousity?

The importance of modelling choices

- Principled: clear inductive biases, $P(\theta)$.
- Efficient: scalable to larger datasets, $N \gg 1$.
- Free of ad-hoc parameters and heuristics, $\nexists \eta, K, \dots$



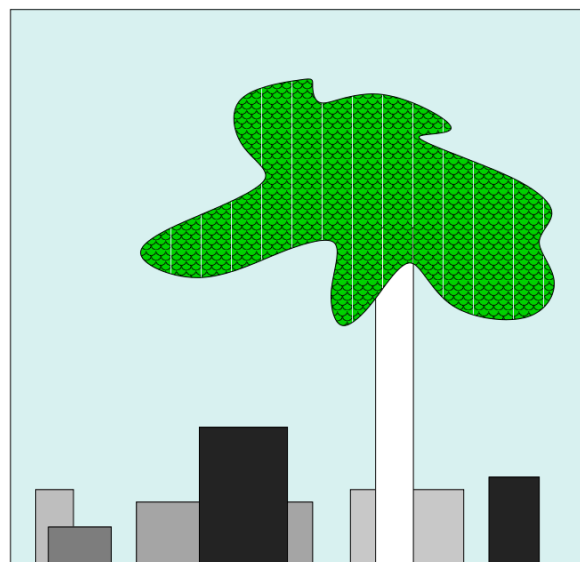
Cabral, Kringelbach, and Deco, 2022



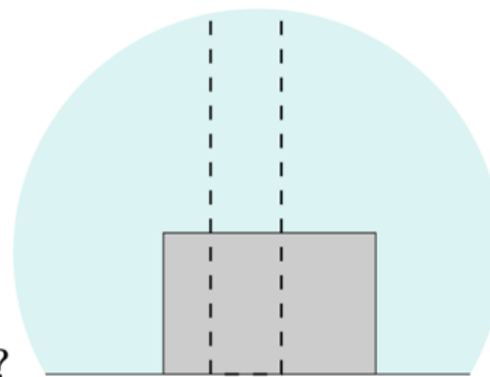
Fortunato, 2016

Occam's razor

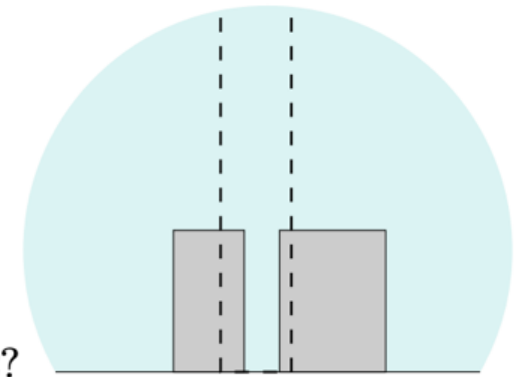
The simplest explanation consistent with observation is the best.



1?



or 2?



How many boxes behind the tree?

MDL principle and network entropies

Best model is that with the shortest description.

$$DL = \text{Info}(\text{Model}) + \text{Info}(\text{Data} \mid \text{Model})$$

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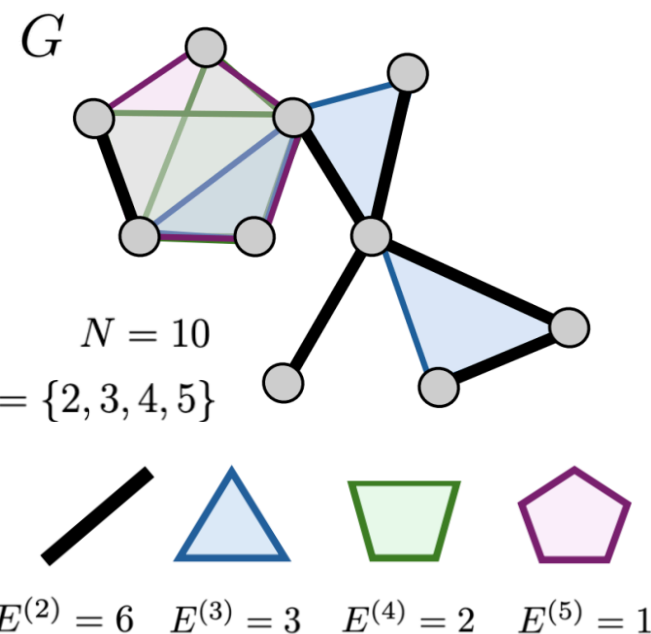
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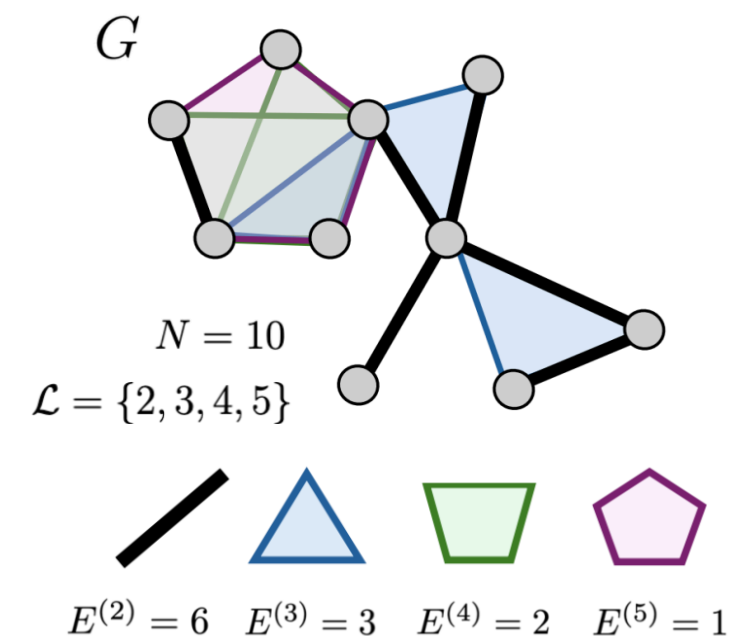
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$$\begin{aligned} H(G) &= \log (\# \text{ possible } G \mid N, \mathcal{L}, \{E^{(\ell)}\}) \\ &= \sum_{\ell \in \mathcal{L}} \log \binom{N}{E^{(\ell)}} \end{aligned}$$

Computing conditional entropies

Amount of information required to transmit G_1
given that G_2 and its overlap with G_1 are known.
“Overlap” is where we impose our inductive biases.

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Align overlap (same order):

$$H_{\text{align}}(G_2^{(\ell)} \mid G_1^{(\ell)}) = \log \begin{pmatrix} E_1^{(\ell)} \\ E_{12}^{(\ell)} \end{pmatrix} \begin{pmatrix} \binom{N}{\ell} - E_1^{(\ell)} \\ E_2^{(\ell)} - E_{12}^{(\ell)} \end{pmatrix}$$

Cross overlap (different order):

$$H_{\text{cross}}(G_2^{(\ell)} \mid G_1^{(k)}) = \log \begin{pmatrix} E_1^{(k \rightarrow \ell)} \\ E_{12}^{(k \cap \ell)} \end{pmatrix} \begin{pmatrix} \binom{N}{\ell} - E_1^{(k \rightarrow \ell)} \\ E_2^{(\ell)} - E_{12}^{(k \cap \ell)} \end{pmatrix}$$

Hypergraph mutual information

Hypergraph mutual information

Mutual information between G_1 and G_2 is information saved in specifying G_2 when knowing G_1 and their overlap.

$$\text{MI}(G_1; G_2) = H(G_1) - H(G_2|G_1)$$

Hypergraph mutual information

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Normalized mutual information for hypergraphs under encoding c .

$$\text{NMI}_c(G_1, G_2) = \max \left\{ \frac{\text{MI}_c(G_1; G_2)}{H_c(G_2)}, \frac{\text{MI}_c(G_2; G_1)}{H_c(G_1)} \right\}$$

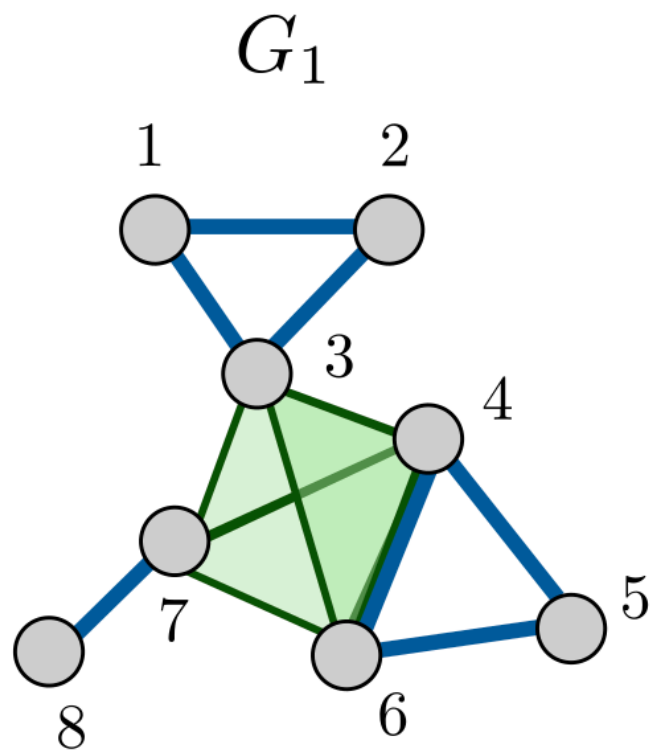
$$H_{\text{align}}(G_2 | G_1) = \sum_{\ell \in \mathcal{L}_2} H(G_2^{(\ell)} | G_1^{(\ell)})$$

(restrict transmission to same-order layers)

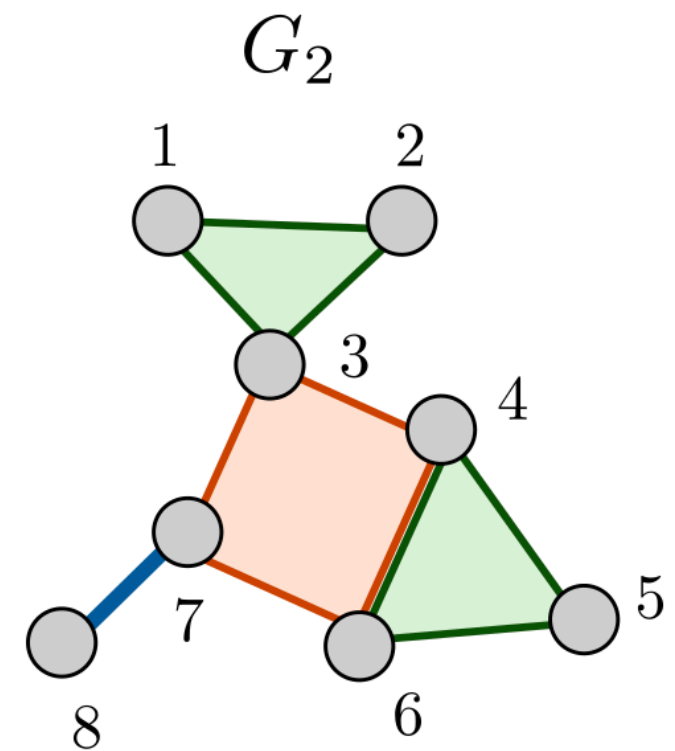
$$H_{\text{cross}}(G_2 | G_1) = \sum_{\ell \in \mathcal{L}_2} H(G_2^{(\ell)} | G_1^{(r(\ell) \rightarrow \ell)})$$

(allow transmission to different-order layers)

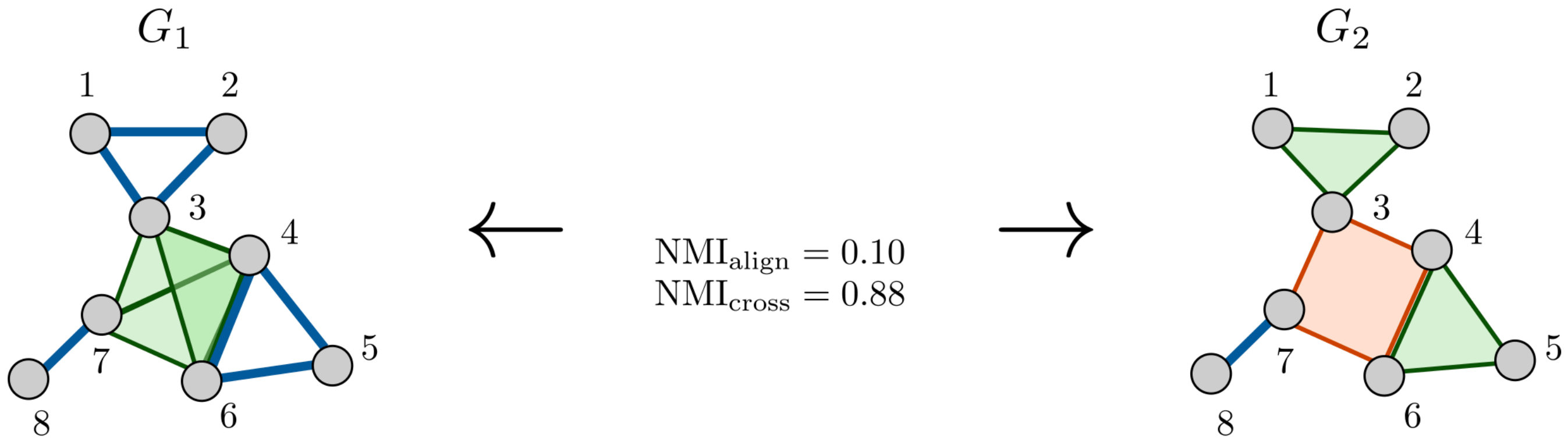
Hypergraph similarity



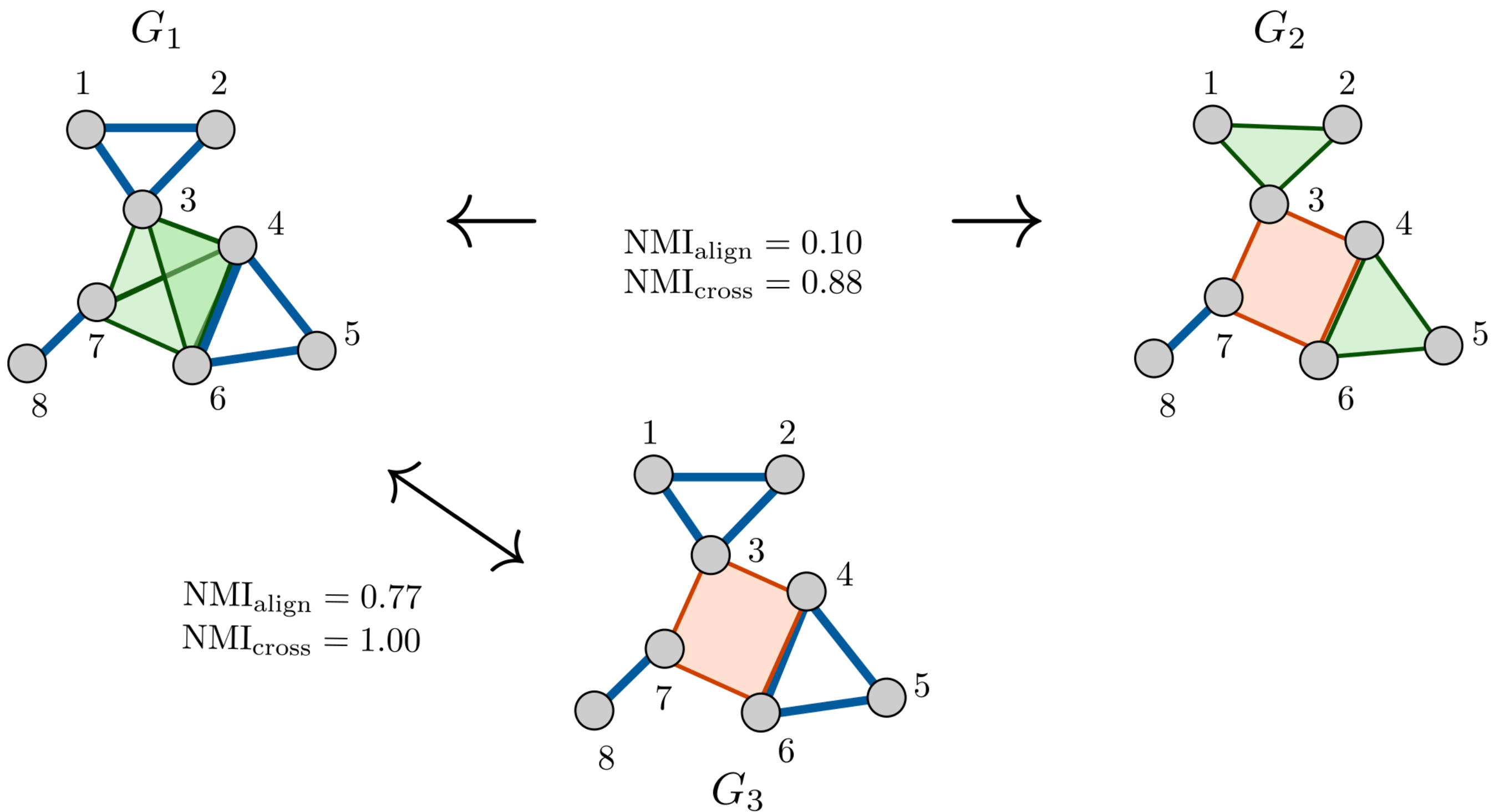
$\stackrel{?}{=}$



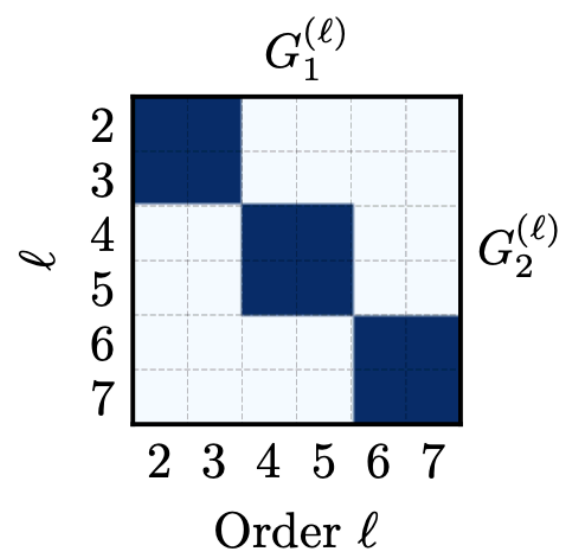
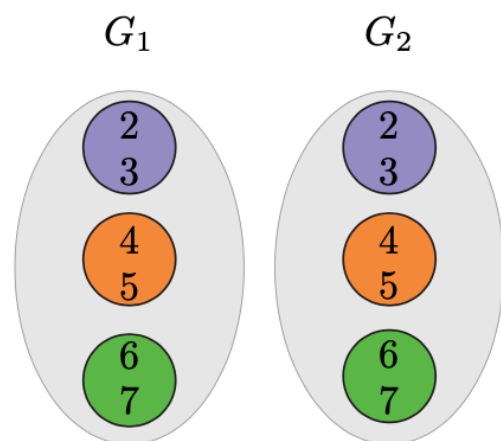
Hypergraph similarity



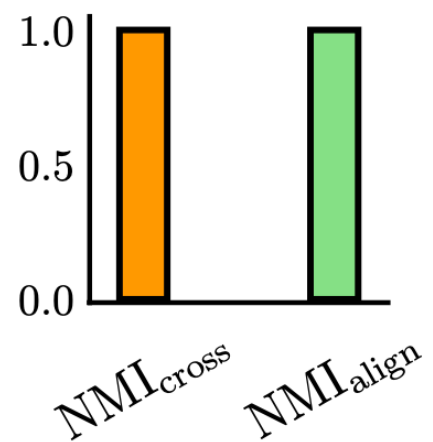
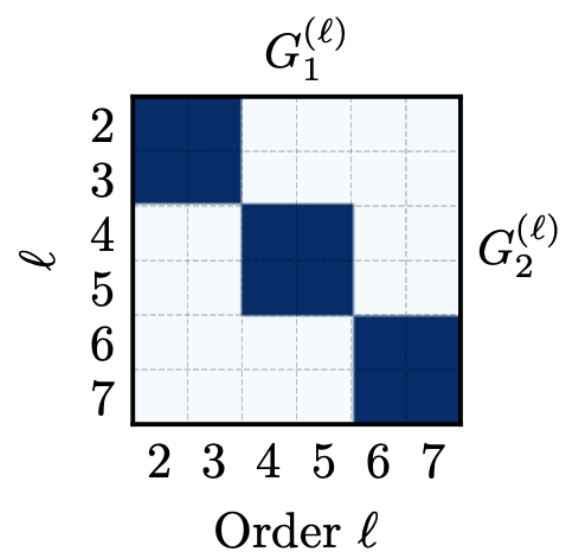
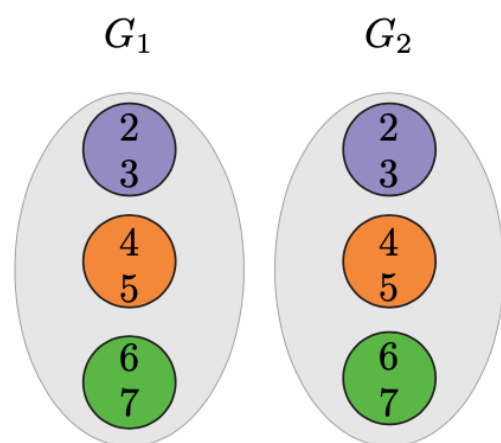
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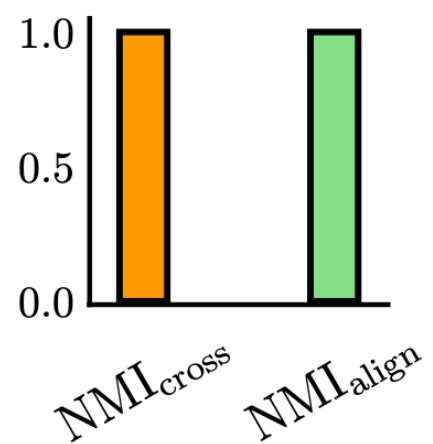
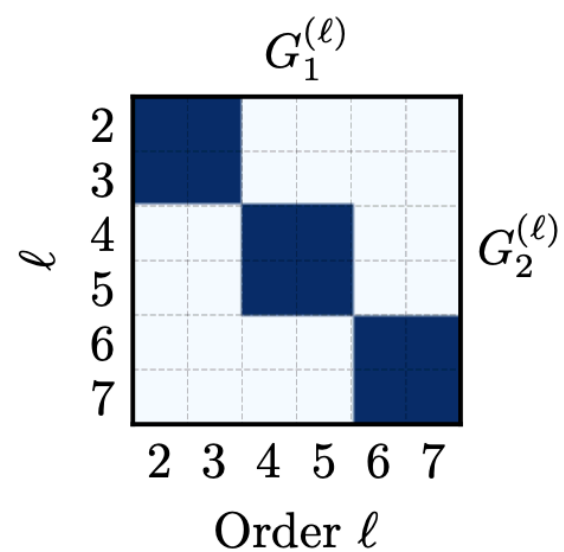
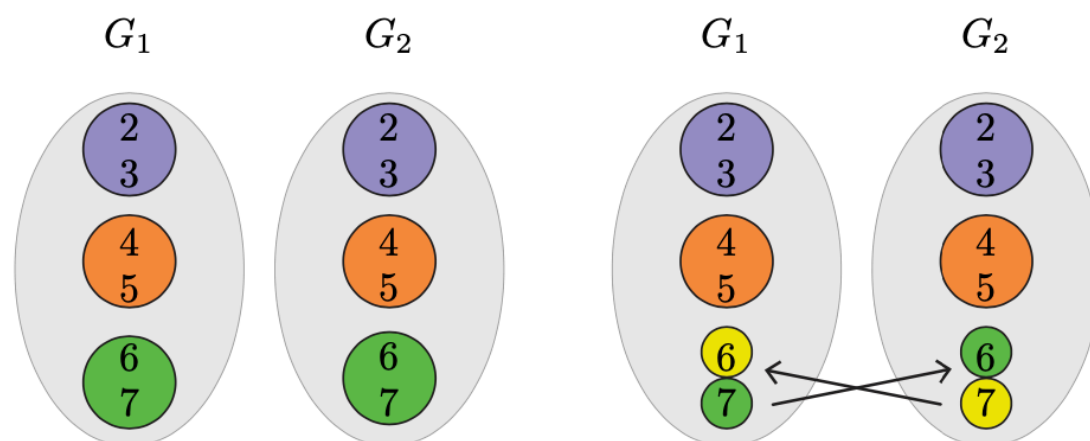
Intra vs. inter



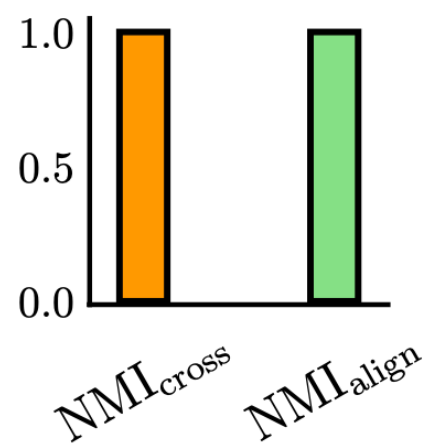
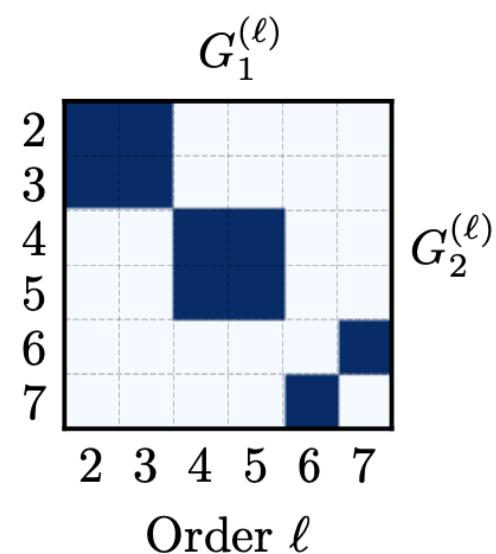
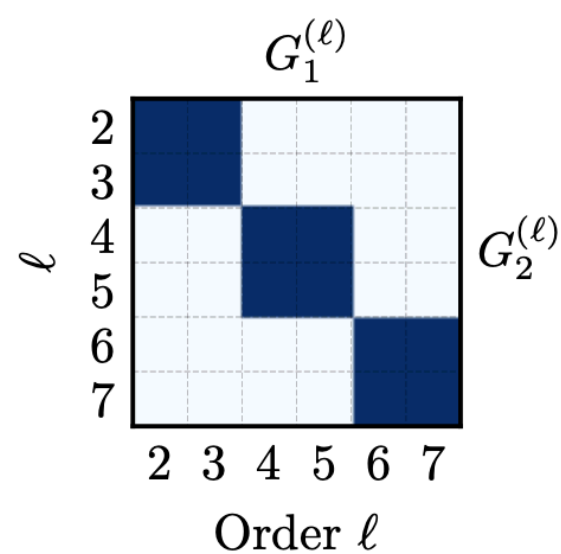
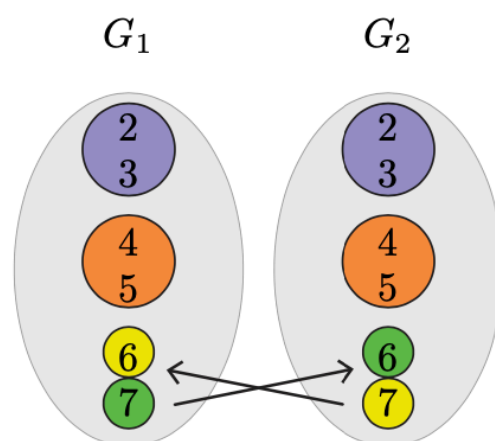
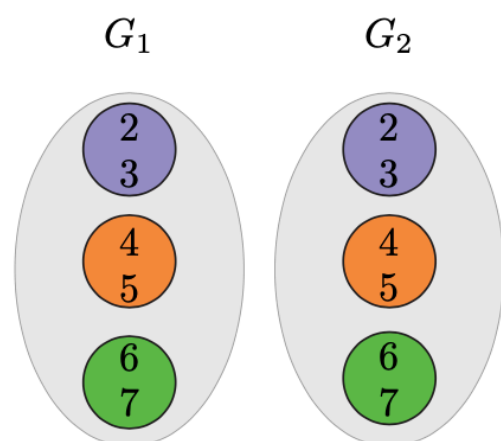
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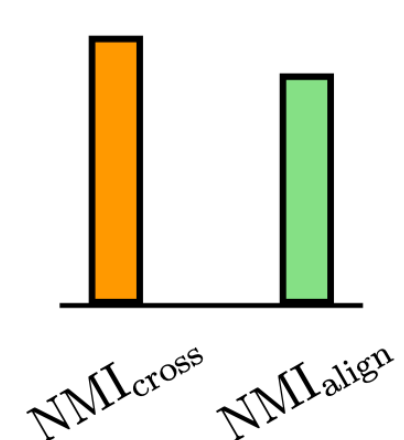
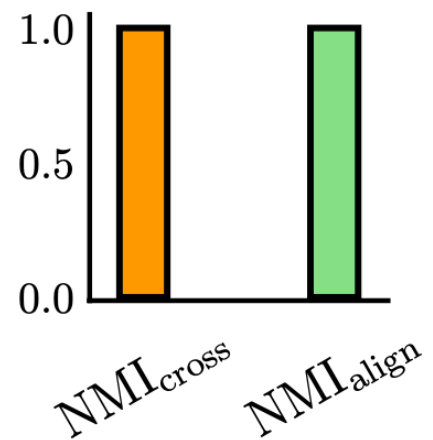
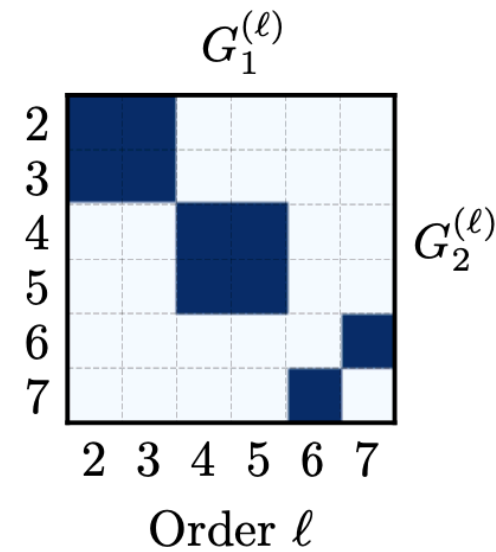
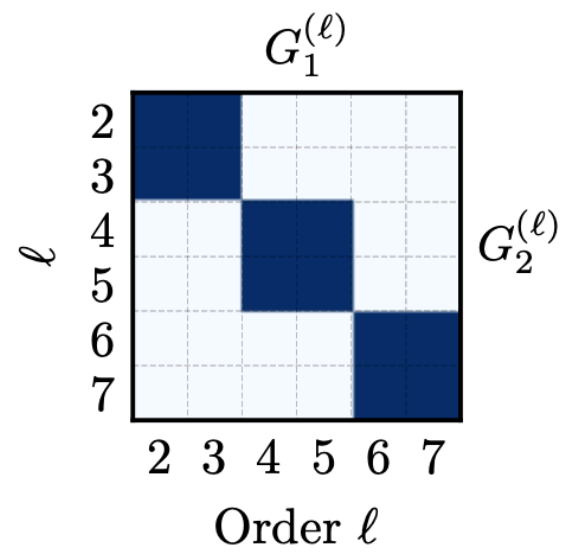
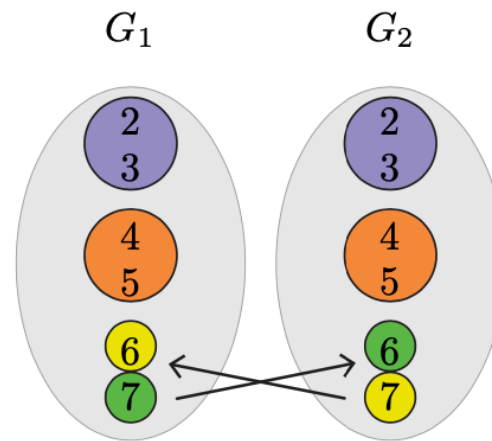
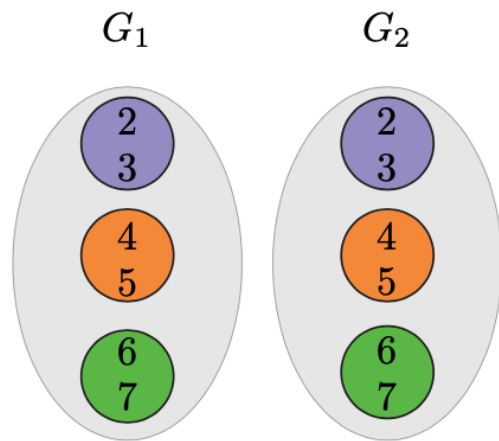
Intra vs. inter



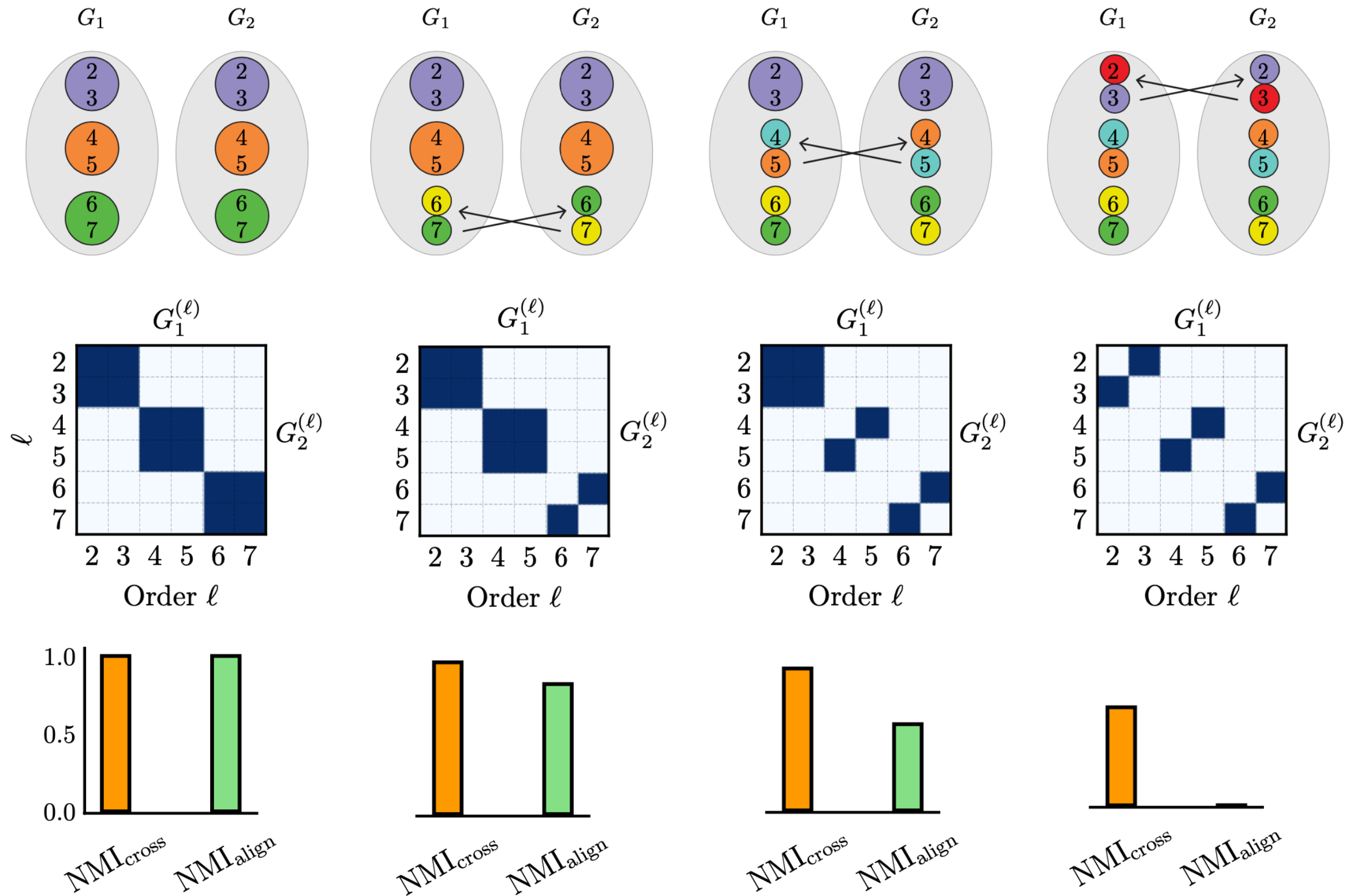
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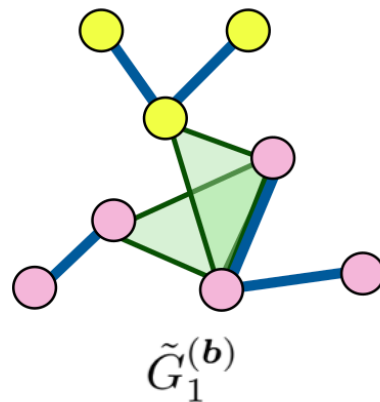
Multiscale hypergraph similarity

Incorporating a partition to the framework.

$$H_{\text{cross}}^{(b)}(G_j|G_i) = \sum_{\ell \in \mathcal{L}} \log \left(\frac{E_i^{(k_i^{(\ell,b)} \rightarrow \ell)}}{E_{i \rightarrow j}^{(k_i^{(\ell,b)} \rightarrow \ell, b)}} \right) \left(\left(\begin{pmatrix} B \\ \ell \end{pmatrix} \right) \left(E_j^{(\ell)} - E_{i \rightarrow j}^{(k_i^{(\ell,b)} \rightarrow \ell, b)} \right) \right)$$

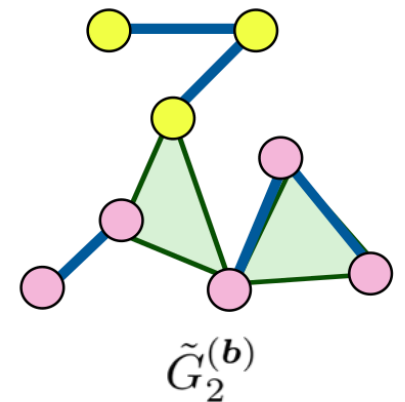
$$H_{\text{cross}}^{(b)}(G_i) = \sum_{\ell \in \mathcal{L}} \log \left(\left(\begin{pmatrix} B \\ \ell \end{pmatrix} \right) E_i^{(\ell)} \right)$$

(analogous formulation to the encoding $c = \text{align}$)



$$\text{NMI}_{\text{cross}} = 0.2$$

$$\text{NMI}_{\text{cross}}^{(b)} = 1.0$$



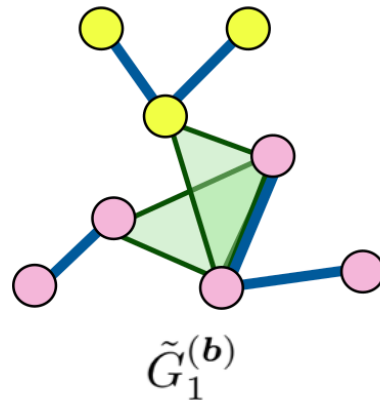
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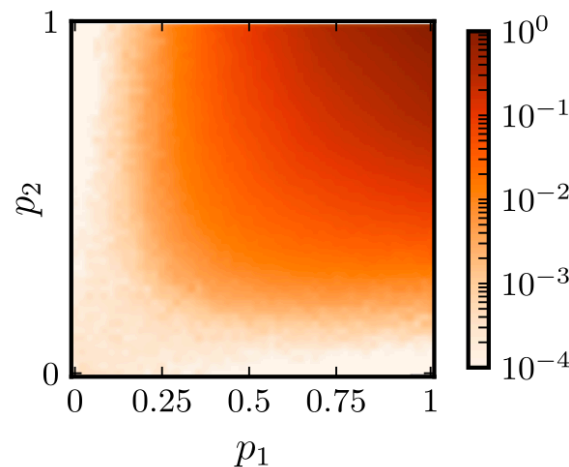
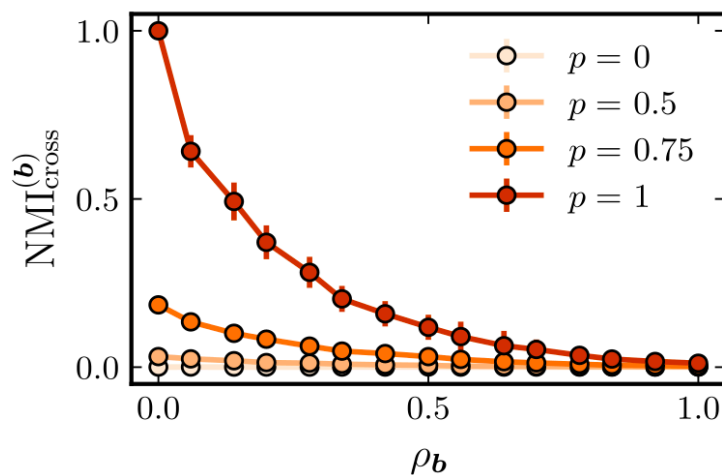
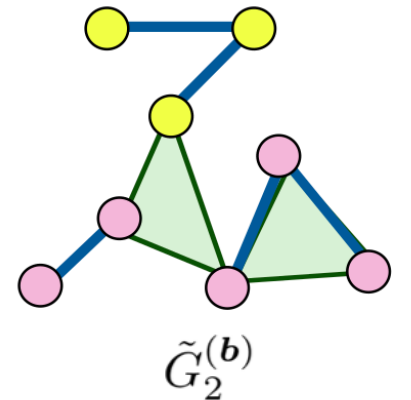
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(analogous formulation to the encoding $c = \text{align}$)



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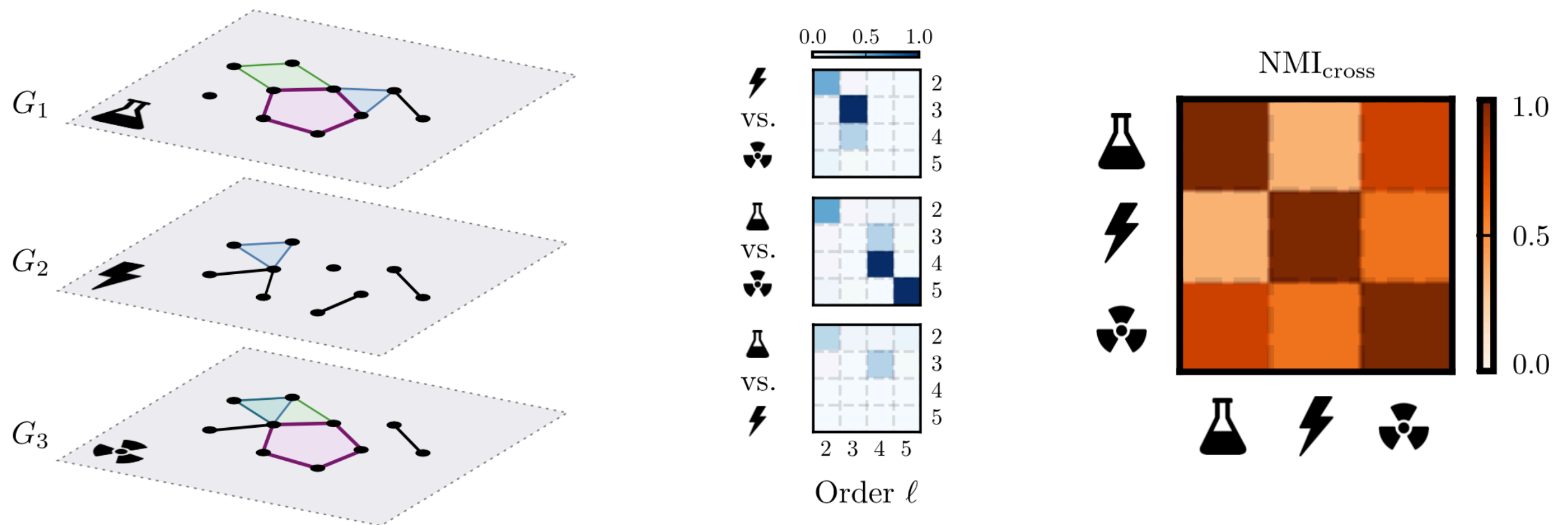
$$\text{NMI}_{\text{cross}}^{(b)} = 1.0$$



(similarity decreases smoothly with noise between underlying partitions)

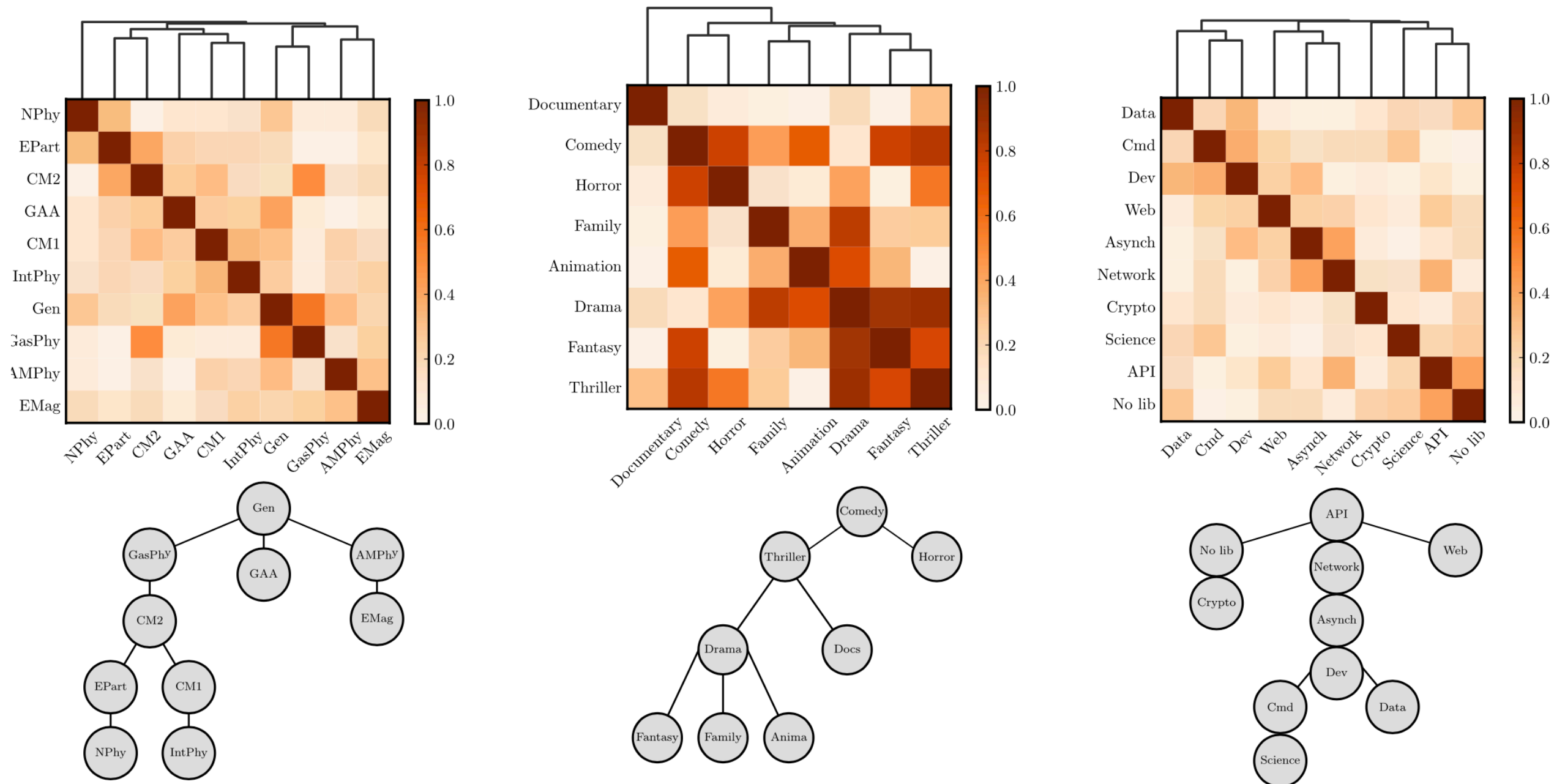
Real-world multiplexes hypergraphs

Application to collaborator networks (APS, IMDb, Git).



Real-world multiplexes hypergraphs

Application to collaborator networks (APS, IMDb, Git).



Summary

Principled, efficient, and interpretable framework for comparing hypergraphs.

Future directions

Extensions to degree-correction and negative non-integer weight encodings.

Adapt method to non-aligned hypergraphs.

Thanks!

H. Felipe, A. Kirkley, F. Battiston. Information theory for hypergraph similarity. *Science Advances* **12**, 23 (2026).



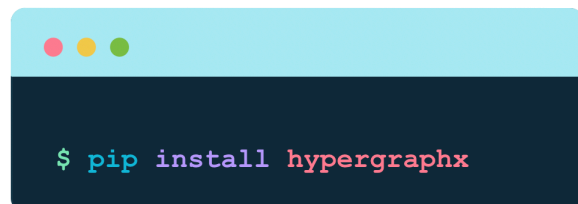
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NETWORK SCIENCE

Information theory for hypergraph similarity

Helcio Felipe^{1†}, Alec Kirkley^{2,3,4*†}, Federico Battiston^{1,5*}

Data and code available as part of the library Hypergraphx.



Code



Data

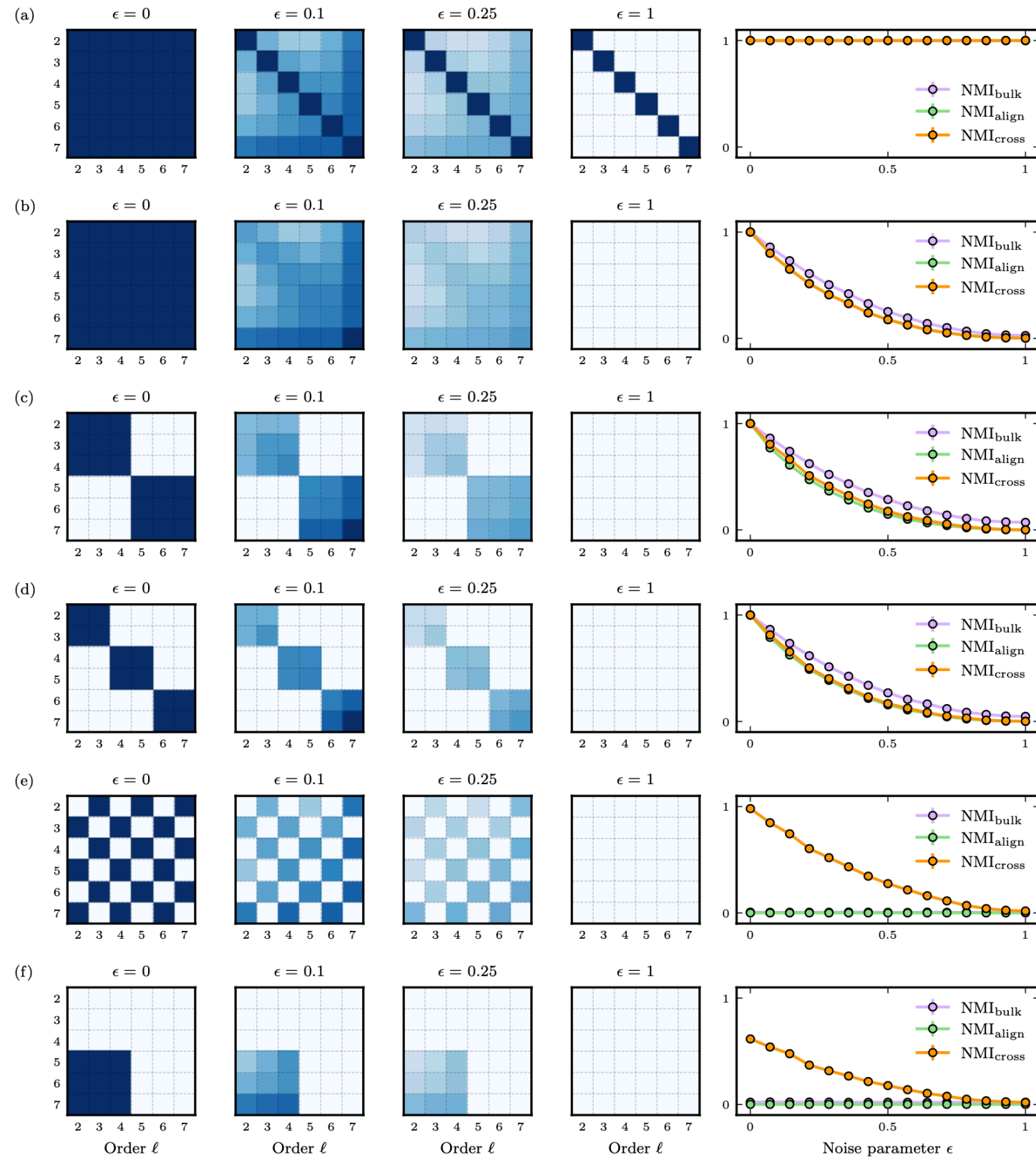


Alec Kirkley
akirkley@hku.hk



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Back-up slides



Back-up slides

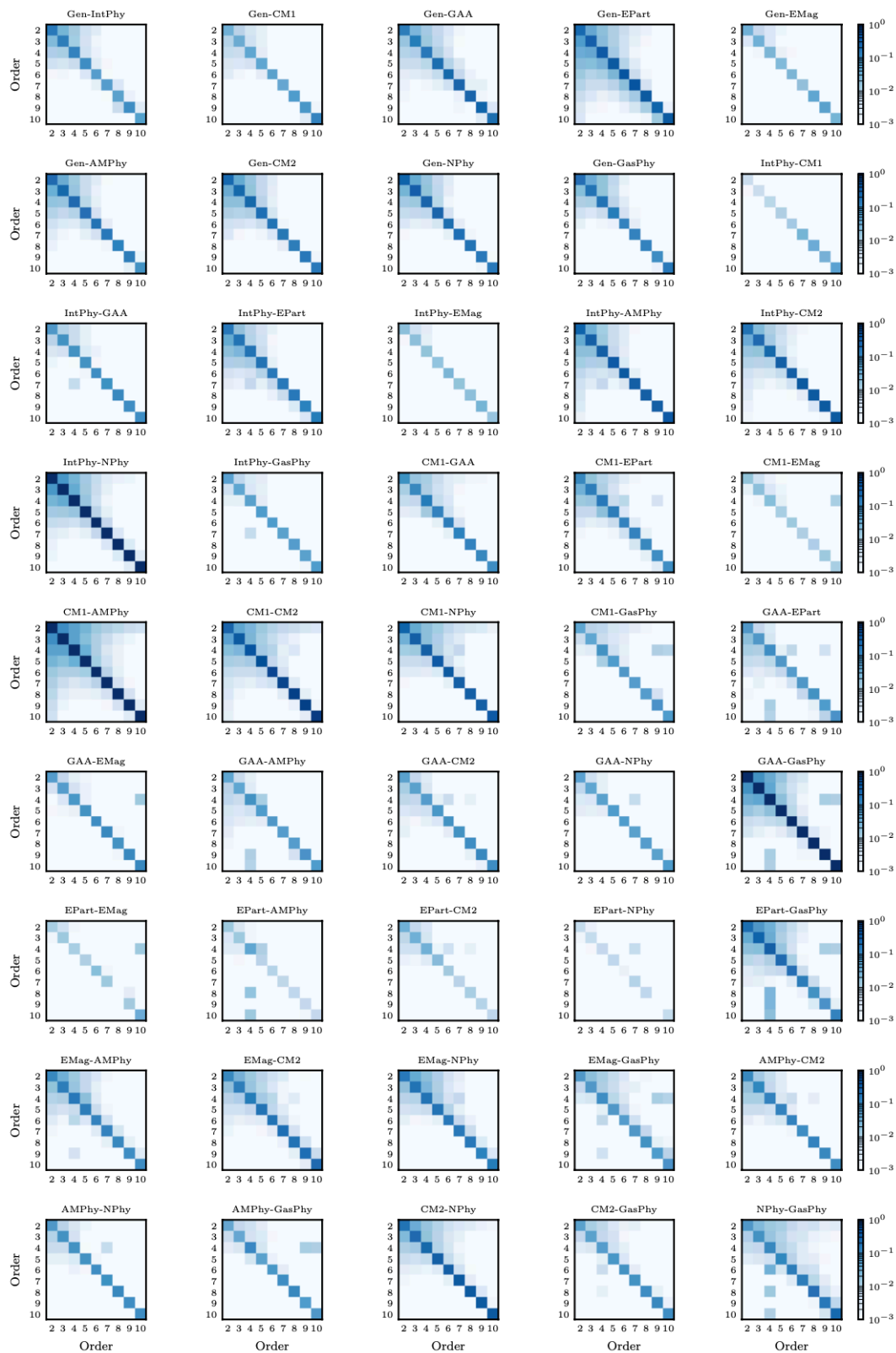


FIG. S5. Order-order similarity matrices of the APS physics fields dataset.

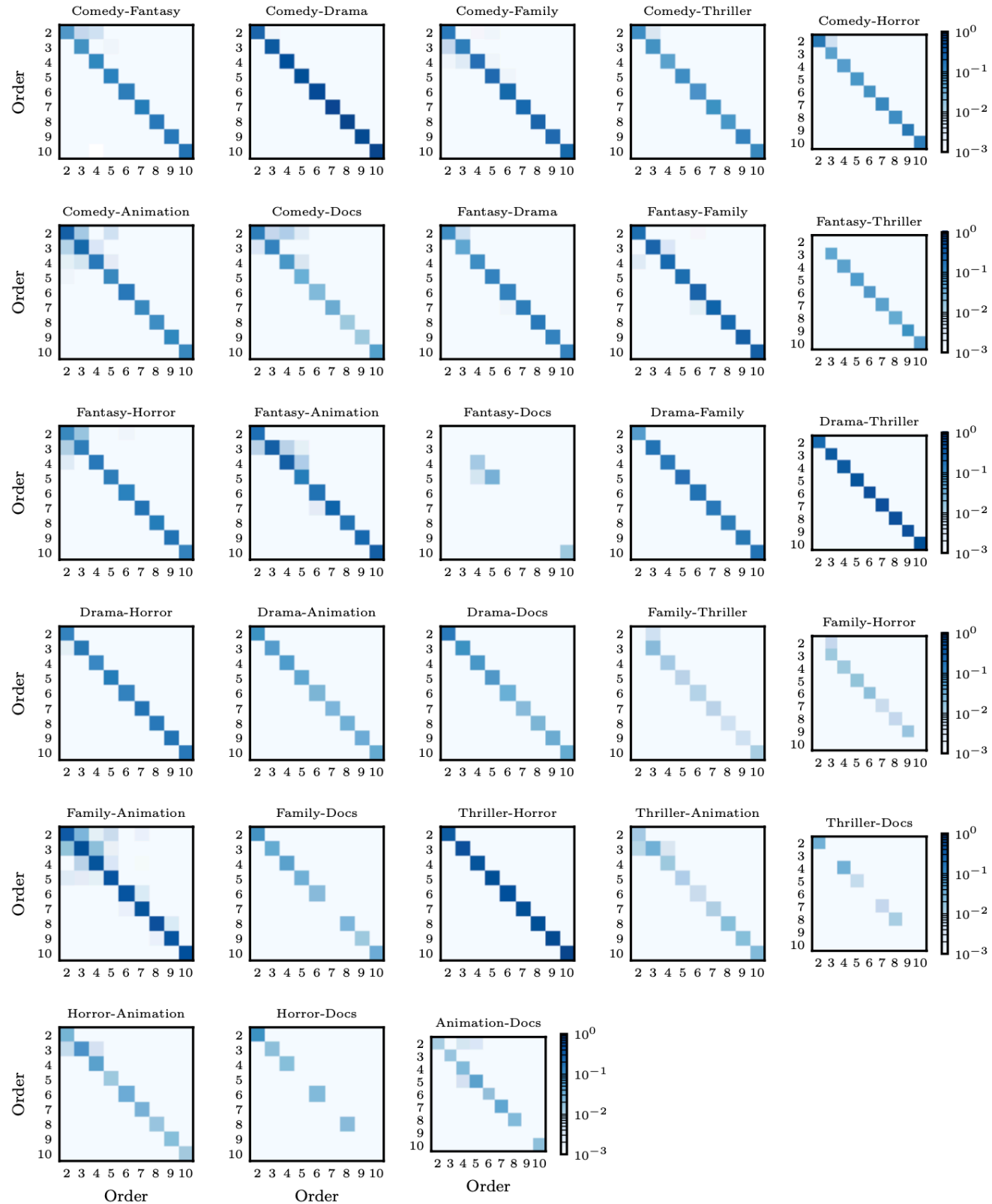


FIG. S6. Order-order similarity matrices of the IMbd movie genres dataset.

Back-up slides

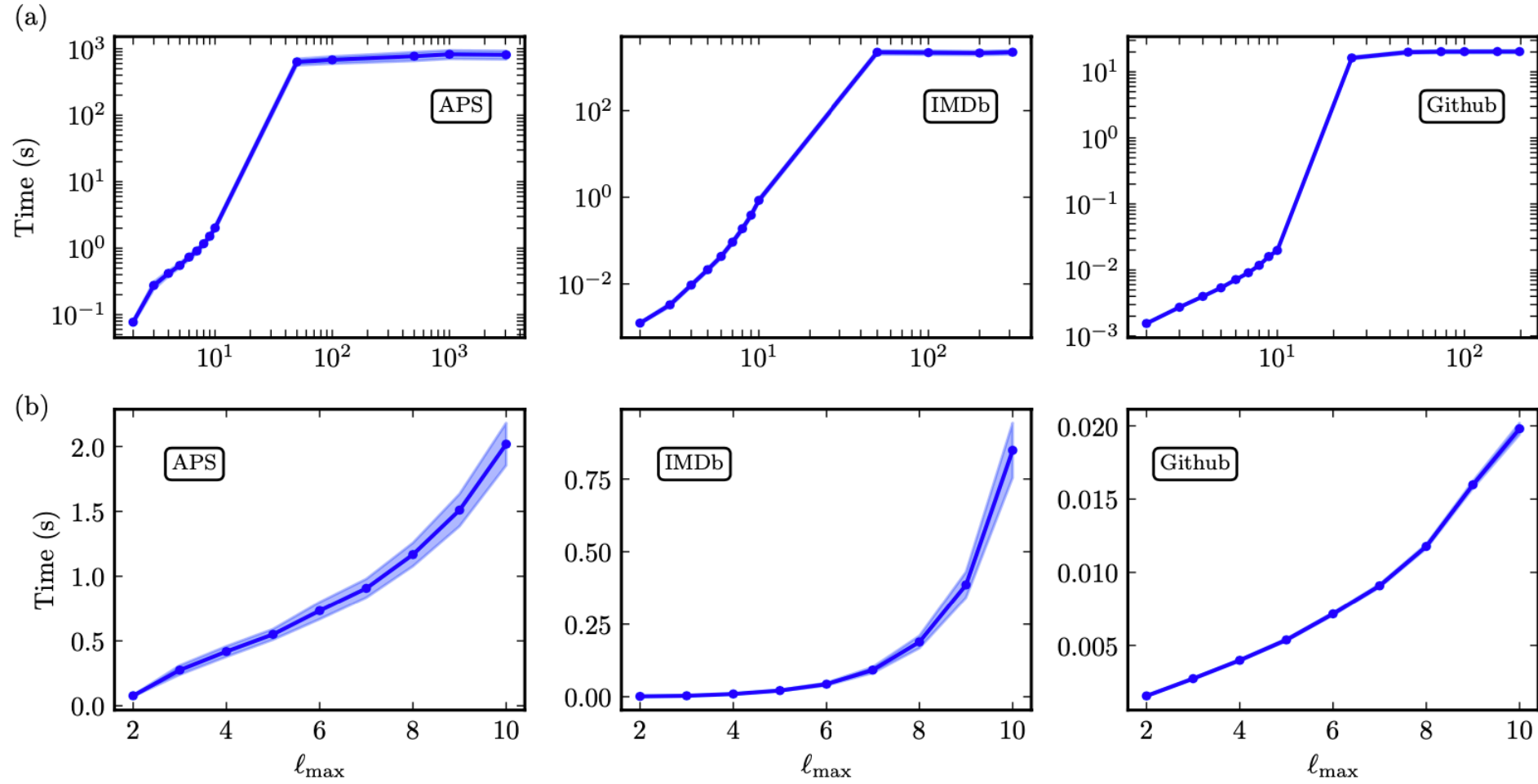


FIG. S8. Runtime scaling in real multiplex data.

Back-up slides

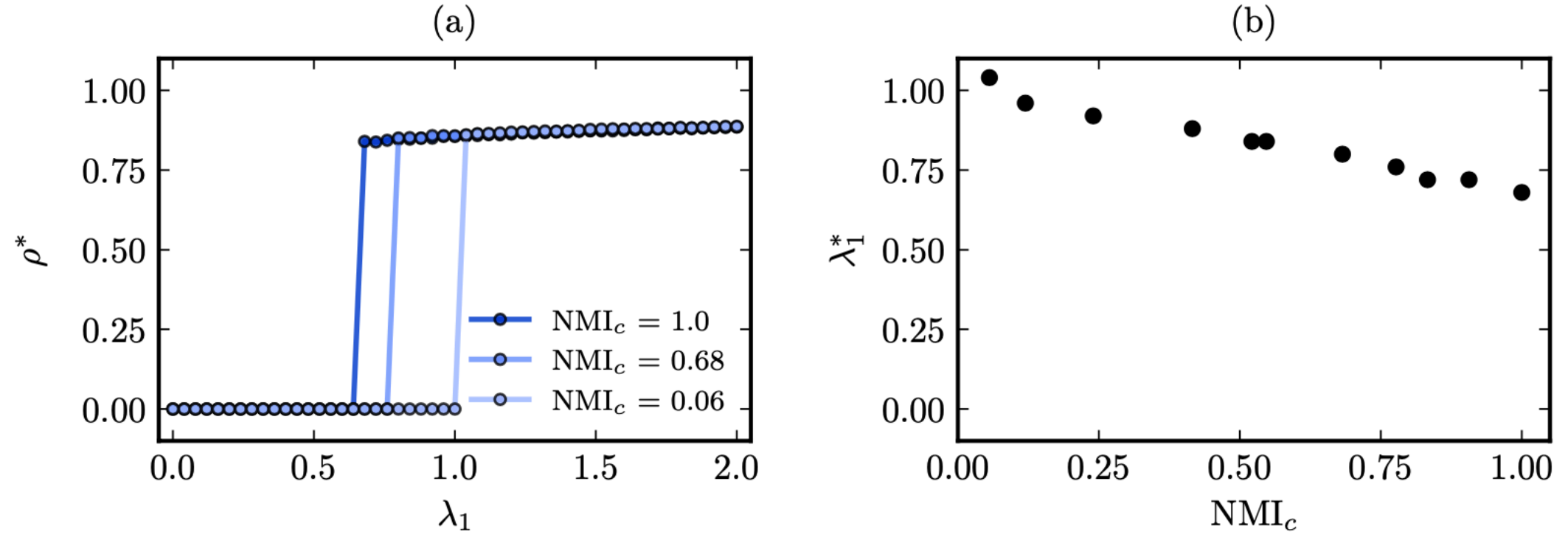


FIG. S10. **Epidemic onset and hypergraph similarity.** (a) Stationary prevalence ρ^* versus infectivity λ_1 for different structural perturbations of a nested hypergraph G . The cross-order NMI between G and its perturbed variant G' for different levels of perturbation noise are shown. (b) Epidemic threshold λ_1^* for the perturbed graph G' as a function of its NMI with G .

Back-up slides

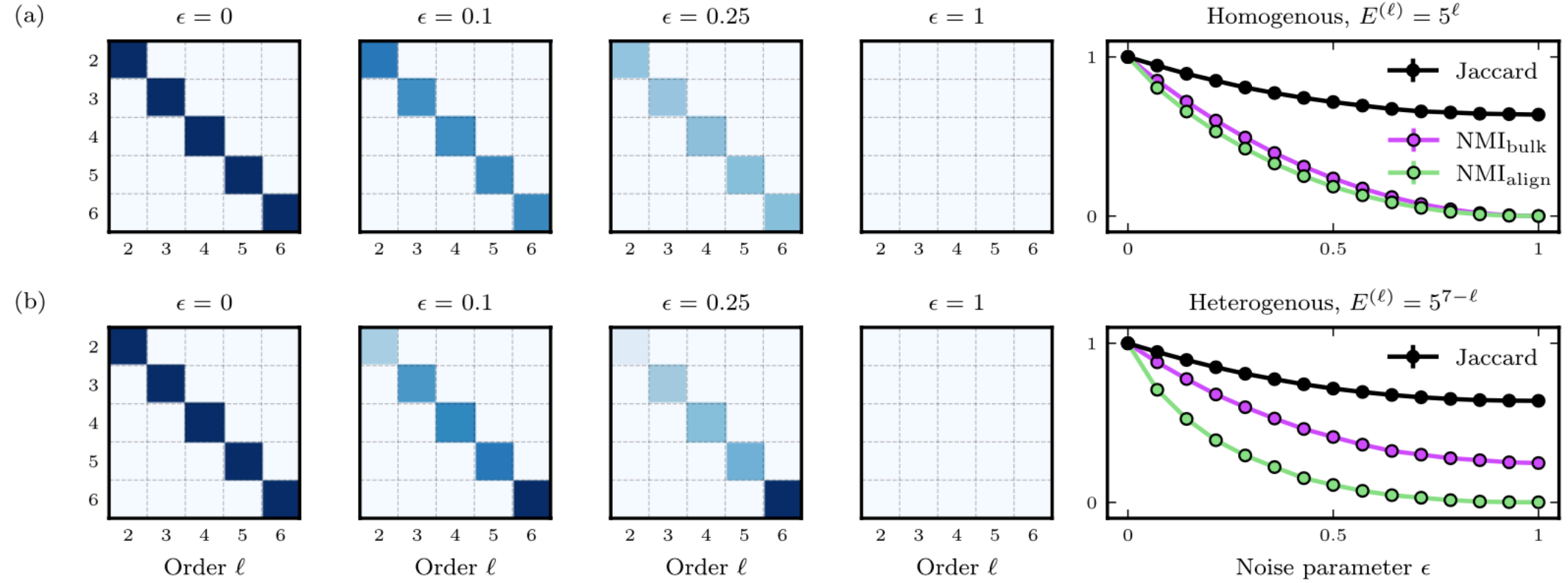


FIG. S11. **Comparison of NMI measures (bulk and align) against the Jaccard similarity index averaged over the best hyperedge matching.** The Jaccard similarity severely inflates the similarity value for uncorrelated hypergraphs, due to spurious hyperedge overlaps attributable to the hyperedge densities.

Back-up slides

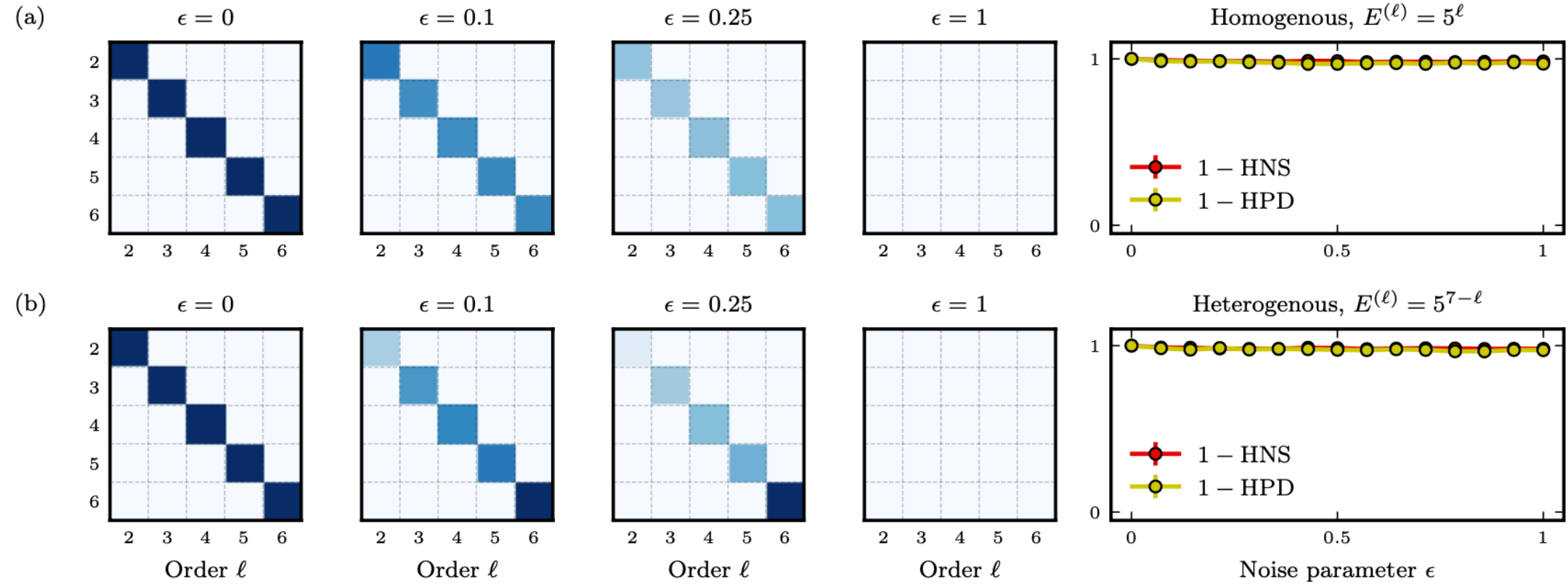


FIG. S12. Synthetic experimental results for the Hyper NetSimile (HNS) and Hyperedge Portrait Divergence (HPD) distance measures, transformed as $1 - \text{distance}$ to form similarity measures.

Back-up slides

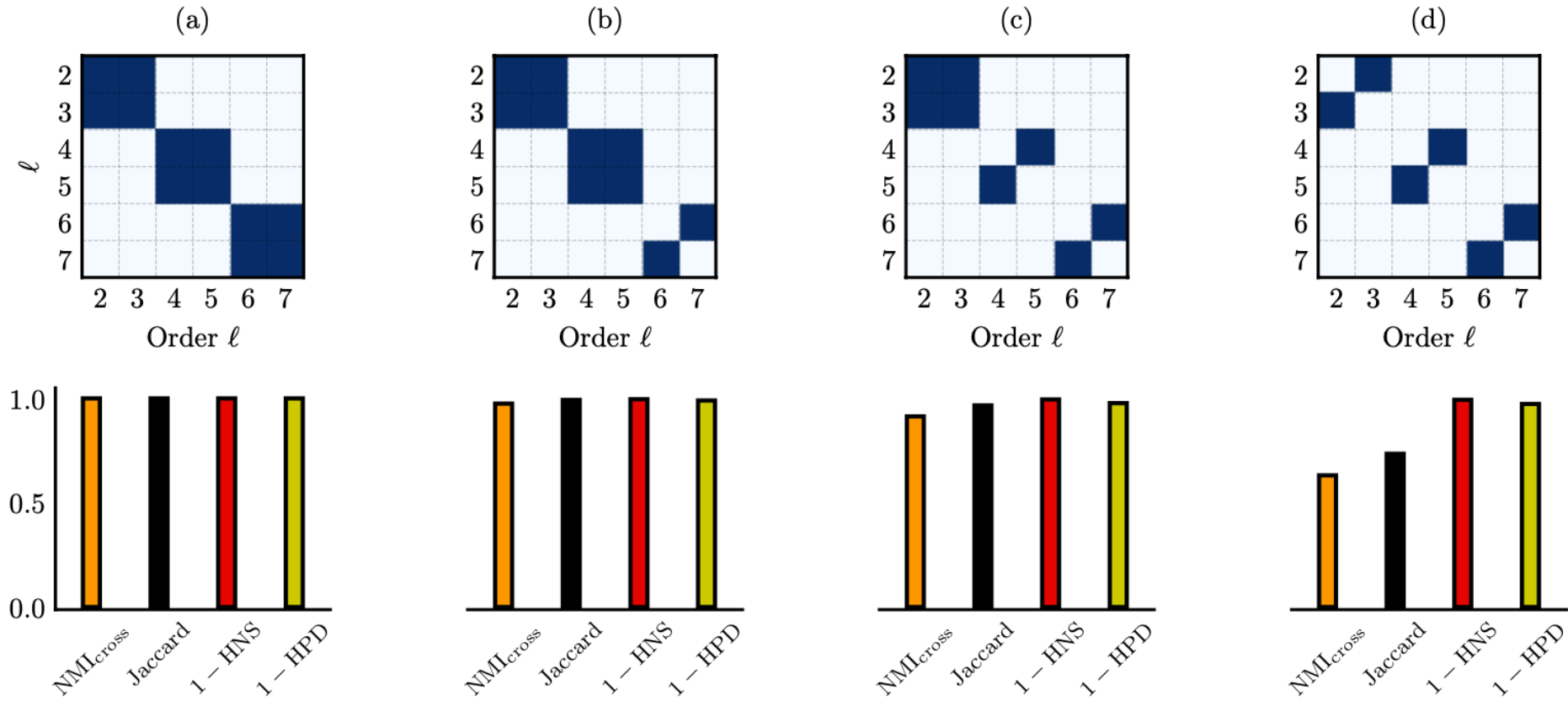


FIG. S13. Comparison of $\text{NMI}_{\text{cross}}$ with the Jaccard baseline and measures of (28) in the block-randomized hypergraph experiment of Fig. 3 in the main text.